



RESEARCH DESIGN · FOR JUDGES

Pilot & Measurement Plan

The research design behind FUTUREPROOF

แผนการนำร่องและการวัดผลกระทบบ · ภาคการศึกษา ๑ / ๒๕๖๙

Dr. Payungsak Kaenchan · Faculty of Liberal Arts, Mahidol University

Submitted to the SPU Tech Creative Learning Awards 2569



Dr. Payungsak Kaenchan | Faculty of Liberal Arts, Mahidol University
Creative Commons BY-NC · Educational, non-commercial use only · No profit · Reuse or duplication must credit the author and obtain prior consent.

Pilot & Measurement Plan

แผนการนำร่องและการวัดผลกระทบ · ภาคการศึกษา ๑ / ๒๕๖๙

This submission is a launch-ready innovation in its zero-cohort phase. Other submissions show what happened — we show what we are prepared to measure. The data study is the next phase, not the current one.

The pilot · semester 1 / academic year 2569

FIRST COHORT	LALA109 · Sem 1 / AY 2569 · August 2026 · Mahidol University, Faculty of Liberal Arts
COHORT SIZE	Up to 50 learners per section × up to 6 sections = up to 300 learners
DURATION	15 weeks · 3 in-class hours + 6 async hours per week (135 hours total)
COURSE FIT	An add-on that runs alongside the existing Design-Thinking final project across the whole semester – students build SDG knowledge in Thai contexts here while developing that project. It augments the course; it replaces nothing.
TEACHER PREPARATION	10 hours pre-term · workshop + Teacher's Manual self-study

What we measure · five instruments

INSTRUMENT	CAPTURES	METHOD
Rubric A · capstone	Voice for Change argument + craft	Teacher · summative · 5 dimensions × 1.0/0.5 · max 45 pts
Rubric B · in-mission	Decision quality across 5 stages	AI · formative · stage-anchored · token-graded
Rubric C · CEFR delta	Language gain (pre / mid / post)	5 dimensions × half-band increments · target +4 half-bands
Rubric D · soft skills	Critical · analytical · collaboration · ethics ·	Teacher + peer · 5 dimensions × max 6 pts each



	decision	
Attitude shift	Pre / post attitudes toward SDGs · AI	Likert-scale survey · n = full cohort

Reliability & ethics safeguards

- **Inter-rater reliability** · Cohen's $\kappa \geq 0.7$ target between two human evaluators on a 10% audit sample.
- **AI evaluator calibration** · random 5% audit against teacher score; drift > 0.5 score-points triggers prompt revision.
- **PDPA compliance** · bilingual consent before any personal-data write; 90-day auto-deletion post-course; granular consent per data category.
- **DPIA on file** · Data Protection Impact Assessment per พ.ร.บ. คุ้มครองข้อมูลส่วนบุคคล พ.ศ. 2562.
- **Withdrawal rule** · learners may withdraw from research participation while continuing the course; their data is purged within 7 days.

Dissemination path · once data lands

Conferences (target)

- TESOL International Convention
- AsiaTEFL Annual Conference
- EuroCALL
- AILA World Congress

Journals (target)

- Computer Assisted Language Learning
- ReCALL
- Language Learning & Technology
- Journal of Digital Learning in Teacher Education

Why this matters · the strategic frame



Past Thai education-innovation competitions have rewarded “evidence of impact” via retrospective classroom data. FUTUREPROOF inverts this: we provide **auditable architecture + a rigorous design ready to produce data**, not retrospective stories from a single uncontrolled term. This is a methodological choice, not a gap.

When the first cohort completes the pilot in December 2026, every instrument above is already built, calibrated, and waiting – so the evidence, when it lands, will be clean, comparable, and publishable.

